Experiment 10: K-Means Implementation

Aim:
To implement K-Means clustering.

Procedure:

1. Import libraries.
2. Load dataset.
3. Apply K-Means.
4. Analyze clusters.

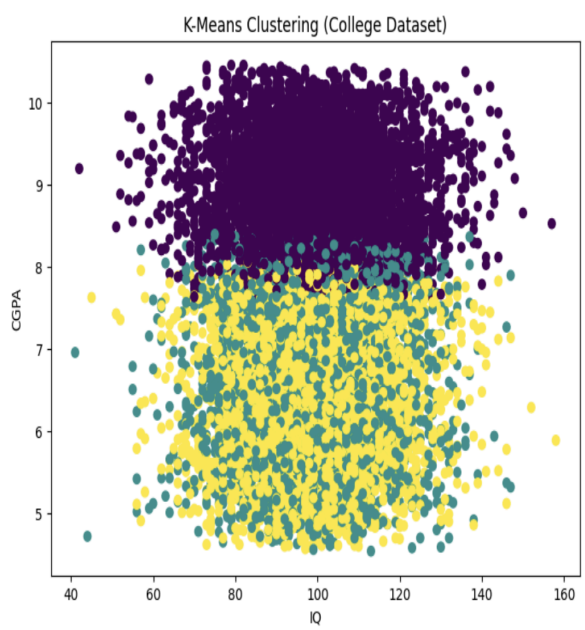
Program:

```
# Experiment 5: K-Means Clustering using college.csv

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Load dataset
df = pd.read_csv("college.csv")

print("First 5 Rows:")
print(df.head())

# Convert Internship_Experience (Yes/No) to 1/0
df["Internship_Experience"] = df["Internship_Experience"].map({"Yes": 1, "No": 0})

# Select numeric features only
features = df.select_dtypes(include=["int64", "float64"])

print("\nFeatures Used for Clustering:")
print(features.columns)

# Scaling (IMPORTANT for KMeans)
scaler = StandardScaler()
scaled_data = scaler.fit_transform(features)
```

```
# Apply KMeans
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(scaled_data)

# Add cluster labels
df["Cluster"] = kmeans.labels_

print("\nClustered Data (First 5 Rows):")
print(df.head())

# Visualization (using CGPA & IQ for 2D plot)
plt.figure(figsize=(8,6))
plt.scatter(df["IQ"], df["CGPA"], c=df["Cluster"])
plt.xlabel("IQ")
plt.ylabel("CGPA")
plt.title("K-Means Clustering (College Dataset)")
plt.show()
```

Output:

```
First 5 Rows:
  College_ID   IQ  Prev_Sem_Result  CGPA  Academic_Performance  \
0    CLG0030  107             6.61  6.28                     8
1    CLG0061   97             5.52  5.37                     8
2    CLG0036  109             5.36  5.83                     9
3    CLG0055  122             5.47  5.75                     6
4    CLG0004   96             7.91  7.69                     7

  Internship_Experience  Extra_Curricular_Score  Communication_Skills  \
0                    No                       8                     8
1                    No                       7                     8
2                    No                       3                     1
3                   Yes                       1                     6
4                    No                       8                    10

   Projects_Completed Placement
0                    4        No
1                    0        No
2                    1        No
3                    1        No
4                    2        No

Features Used for Clustering:
Index(['IQ', 'Prev_Sem_Result', 'CGPA', 'Academic_Performance',
       'Internship_Experience', 'Extra_Curricular_Score',
       'Communication_Skills', 'Projects_Completed'],
      dtype='object')

Clustered Data (First 5 Rows):
  College_ID   IQ  Prev_Sem_Result  CGPA  Academic_Performance  \
0    CLG0030  107             6.61  6.28                     8
1    CLG0061   97             5.52  5.37                     8
2    CLG0036  109             5.36  5.83                     9
3    CLG0055  122             5.47  5.75                     6
4    CLG0004   96             7.91  7.69                     7

   Internship_Experience  Extra_Curricular_Score  Communication_Skills  \
0                      0                       8                     8
1                      0                       7                     8
2                      0                       3                     1
3                      1                       1                     6
4                      0                       8                    10

   Projects_Completed Placement  Cluster
0                    4        No        2
1                    0        No        2
2                    1        No        2
3                    1        No        1
4                    2        No        2
```



Result:

K-Means clustering was successfully performed.